

the binary progenitor system (Fig. 2) (see SOM), ruling out the continued presence of the K-supergiant star and, hence, confirming it as the progenitor of SN 1993J. At the current rate of decay of the SN SED, the U- and B-band fluxes will have reached the level of the proposed B-supergiant component by 2012, at which stage it will be possible to directly measure the properties of the remaining companion star.

These results provide observational proof that red supergiant stars are the progenitors of type II SNe, through the disappearance of the previously identified candidate stars of two SNe. Our best estimate for the mass of the progenitor of the type IIP SN 2003gd is $7 M_{\odot}$ which is at the lower end of the mass range considered theoretically possible to produce core-collapse events. Although the uncertainties ($7^{+6}_{-1} M_{\odot}$) would comfortably allow a large mass for this object, it is interesting that five progenitors of type IIP SNe are found with best estimates at $9 M_{\odot}$ or below (2). This limit is predicted by stellar and SN evolution models (23) and is consistent with the upper initial mass limit observed for white dwarfs (24). The confirmation of the disappearance of the K-

type progenitor star of SN 1993J is further evidence that the binary model previously suggested is valid. It demonstrates the importance of binary interactions for the production of hydrogen-poor SNe (25).

References and Notes

1. R. Gilmozzi *et al.*, *Nature* **328**, 318 (1987).
2. S. J. Smartt, J. J. Eldridge, R. M. Crockett, J. R. Maund, *ArXiv*, astro-ph/0809.0403 (2008).
3. A. Gal-Yam, D. Leonard, *Nature* **10.1038/nature07934** (2009).
4. R. Evans, R. H. McNaught, *IAU Circ.* **8150**, 2 (2003).
5. M. A. Hendry *et al.*, *Mon. Not. R. Astron. Soc.* **359**, 906 (2005).
6. S. J. Smartt *et al.*, *Science* **303**, 499 (2004).
7. S. D. Van Dyk, W. Li, A. V. Filippenko, *Publ. Astron. Soc. Pac.* **115**, 1289 (2003).
8. I. M. Hook *et al.*, *Publ. Astron. Soc. Pac.* **116**, 425 (2004).
9. C. Alard, R. H. Lupton, *Astrophys. J.* **503**, 325 (1998).
10. C. Alard, *Astron. Astrophys. Suppl. Ser.* **144**, 363 (2000).
11. A. Henden, *GRB Coord. Network* **1242**, 1 (2002).
12. A. E. Dolphin, *Publ. Astron. Soc. Pac.* **112**, 1383 (2000).
13. J. H. Elias, J. A. Frogel, R. M. Humphreys, *Astrophys. J. Suppl. Ser.* **57**, 91 (1985).
14. A. Pastorello *et al.*, *ArXiv*, astro-ph/0901.2075 (2009).
15. B. E. K. Sugerman *et al.*, *Science* **313**, 196 (2006).
16. J. A. Cardelli, G. C. Clayton, J. S. Mathis, *Astrophys. J.* **345**, 245 (1989).
17. G. Aldering, R. M. Humphreys, M. Richmond, *Astron. J.* **107**, 662 (1994).
18. K. Nomoto *et al.*, *Nature* **364**, 507 (1993).
19. P. Podsiadlowski, J. J. L. Hsu, P. C. Joss, R. R. Ross, *Nature* **364**, 509 (1993).
20. S. E. Woosley, R. G. Eastman, T. A. Weaver, P. A. Pinto, *Astrophys. J.* **429**, 300 (1994).
21. S. D. Van Dyk *et al.*, *Publ. Astron. Soc. Pac.* **114**, 1322 (2002).
22. J. R. Maund, S. J. Smartt, R. P. Kudritzki, P. Podsiadlowski, G. F. Gilmore, *Nature* **427**, 129 (2004).
23. J. J. Eldridge, C. A. Tout, *Mon. Not. R. Astron. Soc.* **353**, 87 (2004).
24. K. A. Williams, M. Bolte, D. Koester, *ArXiv*, astro-ph/0811.1577 (2008).
25. P. Podsiadlowski *et al.*, *Astrophys. J.* **612**, 1044 (2004).
26. J. R. Maund, thesis, Institute of Astronomy, Cambridge (2005), www.dark-cosmology.dk/~justyn/research
27. J.R.M. is a Sophie and Tycho Brahe Fellow.

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SOM Text

Figs. S1 to S4

Tables S1 to S9

References

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Greatly Increased Toughness of Infiltrated Spider Silk

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In nature, tiny amounts of inorganic impurities, such as metals, are incorporated in the protein structures of some biomaterials and lead to unusual mechanical properties of those materials. A desire to produce these biomimicking new materials has stimulated materials scientists, and diverse approaches have been attempted. In contrast, research to improve the mechanical properties of biomaterials themselves by direct metal incorporation into inner protein structures has rarely been tried because of the difficulty of developing a method that can infiltrate metals into biomaterials, resulting in a metal-incorporated protein matrix. We demonstrated that metals can be intentionally infiltrated into inner protein structures of biomaterials through multiple pulsed vapor-phase infiltration performed with equipment conventionally used for atomic layer deposition (ALD). We infiltrated zinc (Zn), titanium (Ti), or aluminum (Al), combined with water from corresponding ALD precursors, into spider dragline silks and observed greatly improved toughness of the resulting silks. The presence of the infiltrated metals such as Al or Ti was verified by energy-dispersive x-ray (EDX) and nuclear magnetic resonance spectra measured inside the treated silks. This result of enhanced toughness of spider silk could potentially serve as a model for a more general approach to enhance the strength and toughness of other biomaterials.

In the field of biomaterials research, Bryan and Gibbs suggested that metals such as Cu and Zn, which are incorporated in an inner protein matrix, might play a role in mechanical hardening of the jaws of the marine polychaete worm *Nereis* (1, 2). Many groups have investigated the correlation between the presence of some metal elements, such as Zn, Mn, Ca, or Cu, accumulated in the cuticles of insects or other organisms (in body parts such as mandibles, stingers, claws, and ovipositors) and their enhanced mechanical properties (particularly stiffness and hardness) (3–15). We demonstrated that

inorganic impurities such as Zn, Ti, and Al can be inserted into biomaterials by the multiple pulsed vapor-phase infiltration (MPI) process (Fig. 1), performed with equipment conventionally used for atomic layer deposition (ALD).

For testing, we selected spider dragline silk from major ampullate silk glands, which outperforms almost any human-made fiber in its combination of tensile strength (σ) and extensibility (ϵ) (its so-called extreme toughness, $\int \sigma d\epsilon$) (16, 17). A number of scientists have claimed a strong relation between Zn accumulated in insect cuticles, such as the mandibles of the leaf-cutter ant

(3, 6), locust (9), and marine polychaete worm *Nereis* (10), and the enhanced hardness and stiffness of these biomaterials. Broomell *et al.* showed that a metal ion (Zn^{2+}) plays a critical role in the mechanical properties of *Nereis* jaws, by nanoindentation measurements performed before and after Zn chelation (12). Therefore, we initially alternately exposed the dragline silk to the vapor of diethylzinc (DEZ) [$\text{Zn}(\text{C}_2\text{H}_5)_2$] and water (H_2O) (table S1) (18) in an ALD reactor and observed a considerable increase of both σ_{max} (the maximum stress) and E_{ini} (the initial Young's modulus) as compared to those of the untreated native (N) dragline spider silk (SS) (SS/N), which had mechanical properties close to those reported in the literature for the *Araneus* spider's dragline silk [σ_{max} (SS/N) $\approx$ 1.1 GPa and E_{ini} (SS/N) $\approx$ 10 GPa] (19) (fig. S2 and table S2). We subsequently observed that dragline silks exposed to $\text{Al}(\text{CH}_3)_3$ [trimethylaluminum (TMA)]/ H_2O or $\text{Ti}(\text{OCH}(\text{CH}_3)_2)_4$ [titanium(IV) isopropoxide (TIP)]/ H_2O pulse pairs showed considerably larger improvements in toughness than silks exposed to the DEZ/ H_2O pair.

An ALD process conventionally leads to thin deposited layers of metal oxide on a fiber (Fig. 1). Therefore, the question arises whether the increased σ_{max} could be considered a result of

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these deposited layers. SS/ Al_2O_3 (dragline silk + Al_2O_3 by a TMA/ H_2O pulse pair) and SS/ TiO_2 (dragline silk + TiO_2 by a TIP/ H_2O pulse pair) fibers were prepared. When those fibers were compared to native dragline silk, σ_{max} and ϵ_{max}

were increased (Fig. 2, A and B). Taking into account that thin Al_2O_3 or TiO_2 films generally fail by brittle fracture (fracture strain ϵ_f on the order of 10^{-3}) (20), it seems that because of the outer Al_2O_3 or TiO_2 coating of the silk, E_{ini} in the

range of $0 < \epsilon < 0.002$ increased from $E_{\text{ini}}(\text{SS/N}) \approx 9.7 \text{ GPa}$ to $E_{\text{ini}}(\text{SS}/\text{Al}_2\text{O}_3) = 53 \text{ to } 68 \text{ GPa}$ or $E_{\text{ini}}(\text{SS}/\text{TiO}_2) = 53 \text{ to } 66 \text{ GPa}$ (table S2). However, the increase of the maximum tensile stress and strain was independent of the contribution of the Al_2O_3 or TiO_2 coating. From these results, it was concluded that both σ_{max} and ϵ_{max} of the native spider dragline silk can be generally increased by MPI, and the contribution of the outer metal oxide layer coating on the fiber to the improvement of its mechanical properties is of minor importance.

The question still remains whether separate individual pulses [water pulse (WP) exposure without a TMA pulse (TP) exposure and vice versa] would also affect the silk properties. We prepared samples with 100 pulses of individual precursor vapors (SS/TP/100 and SS/WP/100, table S1 and fig. S3) under the same processing conditions as the MPI samples (SS/ Al_2O_3 /100). The overall σ_{max} and E_{ini} of both SS/TP/100 and SS/WP/100 were somewhat decreased and, in contrast, ϵ increased only slightly (Fig. 2C). However, the overall toughness was much smaller than that of SS/ Al_2O_3 /100. Pulses of only one reactant (TMA or H_2O) were not effective for the enhancement of the mechanical properties of the silk. Additionally, the exposure times appear to play an

Fig. 1. Schematic of the difference between conventional ALD (thin-film deposition, top) and MPI (thin-film deposition plus chemical modification by metal infiltration, bottom). Conventionally, by multiple pulses of ALD precursors such as TMA (green arrows) and water (H_2O , pink arrows), thin films [such as alumina (Al_2O_3), illustrated by the red shell] are deposited on rigid materials (such as metals) without chemical modification of the bulk. In contrast, in the case of soft materials such as biomaterials or polymers, MPI provides a chemical modification of the bulk in addition to the thin-film deposition.

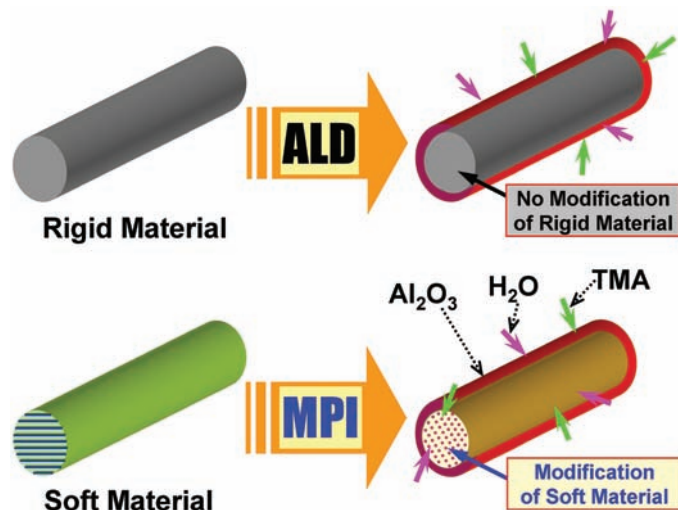
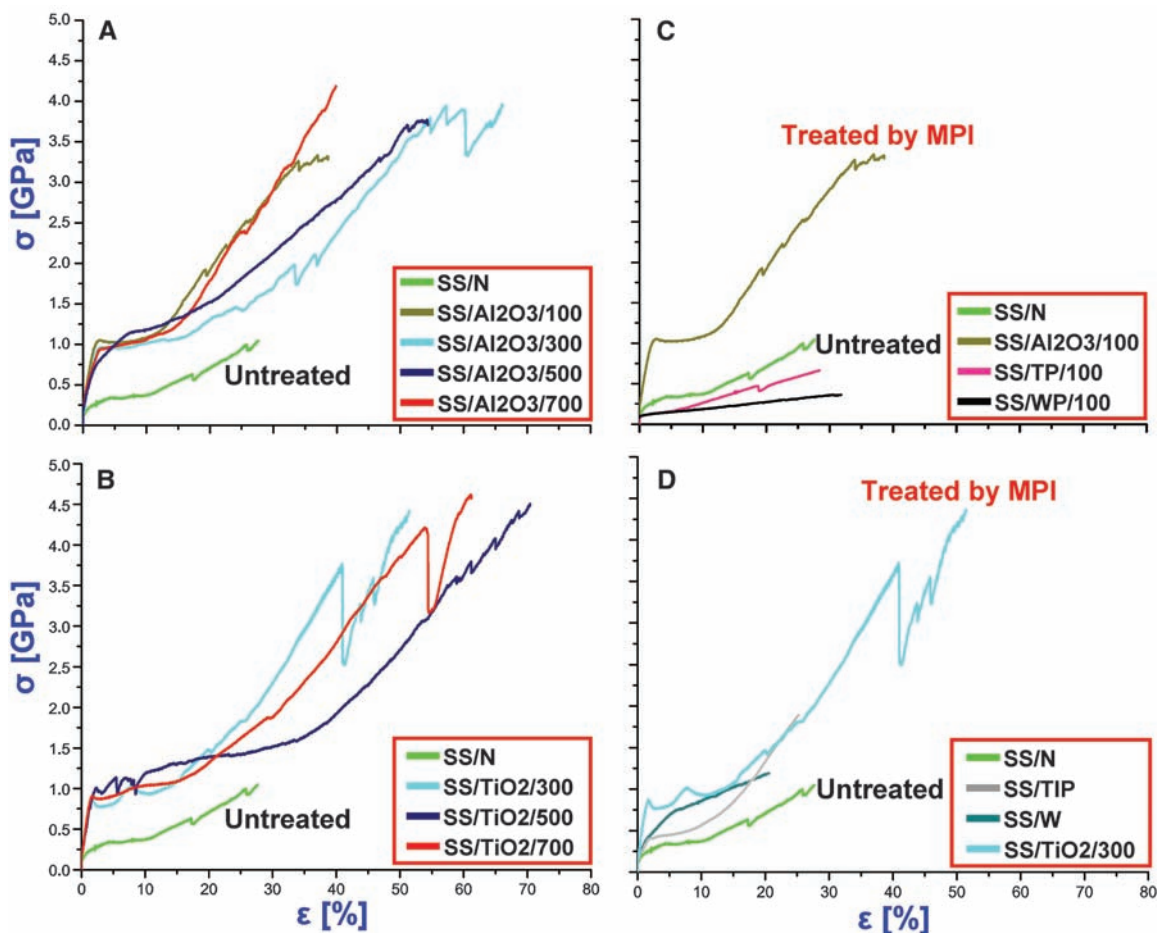
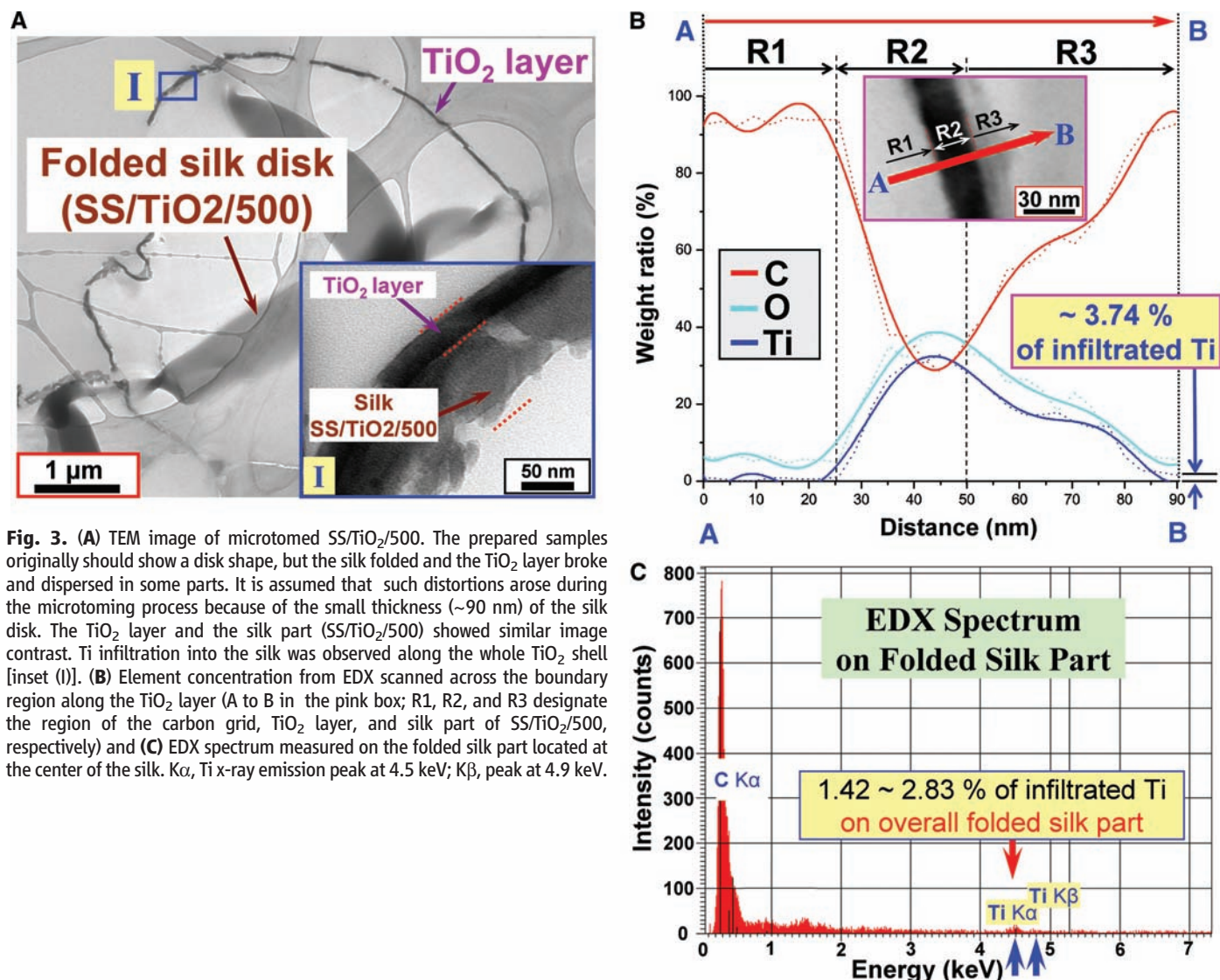


Fig. 2. Tensile test curves of silk fiber samples treated by TMA/ H_2O and TIP/ H_2O pulse exposure pairs and comparison to untreated samples and samples treated with various different processes. (A) and (B) Stress (σ) and strain (ϵ) curves of silk fibers treated with TMA/ H_2O and TIP/ H_2O pulse pairs in various numbers of cycles ranging from 100 to 700. (C) σ - ϵ curves of silk fibers treated by individual vapor pulses (SS/TP/100, TMA without water; SS/WP/100, water without TMA; sample denotation and information can be found in table S1 and see fig. S3) together with a comparative (σ - ϵ) curve of SS/ Al_2O_3 /100. (D) σ - ϵ curves of silks dipped into either a TIP (SS/TIP) or H_2O (SS/W) solution for 10 hours, followed by drying at room temperature, and SS/ TiO_2 /300 as a comparative σ - ϵ curve. In all graphs, the green line indicates the σ - ϵ curve of the native spider silk without any treatment (details are given in table S1).





important role. For the current results, we used a chamber-type ALD setup with exposure times for individual pulses of up to 40 s (see also table S1). Control experiments with a flow-type ALD process, which only allows for an exposure time of less than 1 s, did not yield considerable enhancement of the mechanical properties, clearly indicating a more complex process than simple coating of the structure as would be expected in a regular ALD process (21).

We also investigated whether simple dipping of silks into the highly reactive liquid ALD precursor would enhance the mechanical properties. We prepared various samples that were separately dipped into the individual precursor solutions (SS/TiP, dipped into TiP; and SS/W, dipped into water; table S1) under ambient conditions (at a temperature of 15°C and at atmospheric pressure) for 10 hours, followed by drying under the same conditions (22). E_{ini} and σ_{max} increased and ϵ_{max} decreased slightly as compared to SS/N (Fig. 2D). However, like the samples produced by the individual pulses of only one reactant under ALD

conditions, the overall mechanical properties of those samples (SS/TiP and SS/W) were not comparable to those of SS/TiO₂/300.

The increased toughness of the above-mentioned silks appears to be caused by an infiltration of inorganic impurities such as Al or Ti, which presumably react with proteins after a pre-conditioning by the water penetrating into the fiber (23, 24), a process probably similar to the hardness- and stiffness-increasing effects on *Nereis* jaws produced by a small amount of artificial Zn incorporation into the jaw (12). As shown in transmission electron microscopy (TEM) images of SS/TiO₂/500 (Fig. 3A, inset I, and B), along the TiO₂ shell, a region of ~100 nm in depth shows a high image contrast. Considering the relative weight ratio of carbon (C), oxygen (O), and Ti, a large amount of Ti was infiltrated in this shell region. In the center part of the silks (folded region), energy-dispersive x-ray (EDX) analysis detected weak but clear Ti signals (1.42 to 2.83% by relative weight ratio). Because the resolution limit of the system amounts to about 0.5 to 1.0%,

the spectrum under those limits has not been quantified. But qualitatively, the small amount of Ti shown as the Ti-K peak is well above the background. In addition, indirect but similar evidence for the infiltrated Al ions into silk, which interact with the silk proteins, was observed by magic angle spinning nuclear magnetic resonance (NMR) measurements of SS/Al₂O₃/300 [supporting online material (SOM) text I].

Spider silk is generally regarded as a semi-crystalline biopolymer containing two major parts: β -sheet crystalline parts consisting of hydrophobic polyaniline sequences and amorphous parts composed of amino acid residues, linked via hydrogen bonds (19). The exact infiltration mechanism and intermolecular bonding states between proteins and metals has not been determined. However, considering the severe attack of water occurring at the hydrogen bonds that interconnect the proteins (25), a global weakening of the hydrogen bonds in proteins with increasing water vapor temperature (26) can be assumed. In addition, the strong reactivity to chemical bonds (23) and deep penetrat-

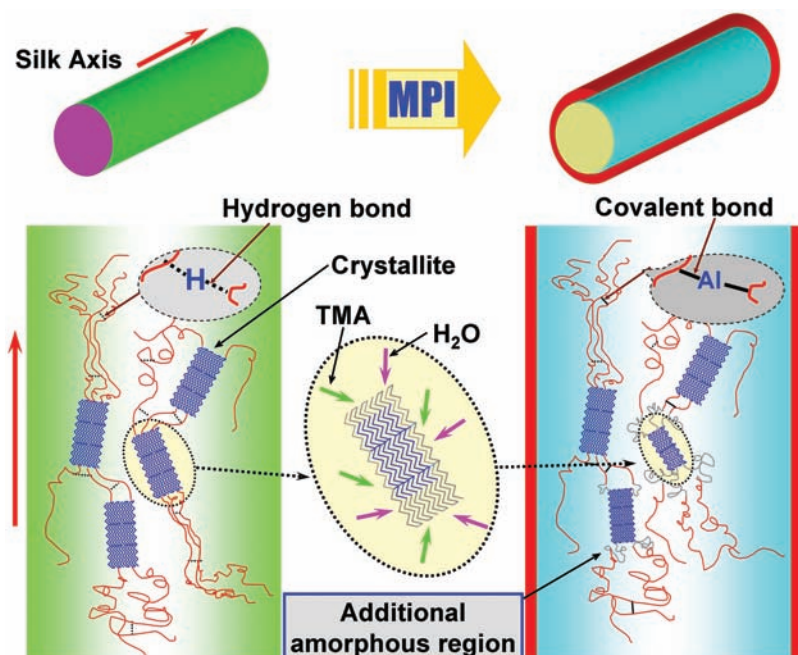


Fig. 4. Schematic description of proposed molecular changes in the silk produced by MPI. The detailed explanation can be found in the text and in fig. S6.

ing capability of metal-containing ALD precursors (such as TMA, TIP, and DEZ) into soft materials such as polymers (24) and the metal ions' preferential binding features to the proteins (27, 28) could lead to stable metal-protein compounds by chelating ions (29, 30) such as Al^{3+} (31), Ti^{4+} (32), or Zn^{2+} (33) to form metal-protein complexes. During long exposure times to water vapor (5 to 40 s) (34), the inner hydrogen bonds of the silk protein are likely to weaken or break in some regions upon water vapor attack at 70°C. Subsequently after long-term exposure to the metal precursor vapor, Al^{3+} , Ti^{4+} , or Zn^{2+} is likely to infiltrate the protein and bind to the broken bonding sites, thereby resulting in the formation of metal-coordinated or even covalent bonds. As a minor additional effect, methane (TMA/ H_2O) or isopropanol (TIP/ H_2O) as reaction byproducts may have additional effects to weaken or break the remaining hydrogen bonds. Consequently, the recoverable hydrogen bonds may be changed to permanent covalently bonded or coordinated Al-, Ti-, or Zn-protein complexes (Fig. 4 and fig. S6). Therefore, unlike native silks, which are highly sensitive to environmental conditions such as humidity and temperature (mainly caused by the hydrogen bond breaking and recovery feature between water and amide groups in the protein chains) (19), the metal-infiltrated silks, which presumably may have covalent or coordinated bonds, are hardly affected by these conditions (35).

By wide-angle x-ray scattering (WAXS) measurements of SS/ Al_2O_3 /300 and SS/ TiO_2 /500, an inner-protein matrix structure change, caused by metal incorporation, was observed. More specifically, the size of silk protein β -sheet crystallites decreased after the MPI treatment (SOM text 2, figs. S4 and S5, and table S3). The strength of the

silk is strongly dependent on the sizes of the polyalanine crystals, which are either small and perfect or large and imperfect (36), as proposed by Termonia (37) and experimentally validated by Du *et al.* (38). The extensibility of the silk is determined by the orientation and amount of amorphous chains, as proposed by Termonia (37) and elucidated by Lefèvre *et al.* (39). From these WAXS results (a decrease in the size of the protein crystallites), it could be easily conjectured that parts of protein crystals (probably large imperfect crystallites) are likely to be converted into protein chains by hydrogen bond breaking and metal substitution caused by MPI, thus resulting in additional amorphous regions. From the theoretical model by Termonia and supportive experimental results by Du *et al.* (38) and Lefèvre *et al.* (39), it is likely that additional amorphous regions and size-reduced protein crystallites, induced by metal infiltration, function as factors to enhance ϵ_{max} and σ_{max} of MPI-treated silks (Fig. 4 and fig. S7). On the other hand, we observed that the individual TMA or H_2O pulses produced no major enhancement of the mechanical properties of the silk composites (Fig. 2C). It appears that even though the individual pulsed vapor molecules contribute to the severe hydrogen bond breaking on the overall silk, because of the strong recovery behavior of the hydrogen bonds, some are reestablished. The resulting silks show an analogous curve profile to those of slightly contracted silk fibers, caused by local hydrogen bond breaking (40). We also observed that direct dipping of the silk into the pure liquid TIP precursors at ambient conditions had only a slight enhancement effect on the mechanical properties (increased σ_{max} of SS/TIP in Fig. 2D), similar to the hardness enhancement of the polychaete worm's

jaws caused by Zn incorporation through incubation in ZnCl_2 solution at 25°C (12). Unlike TIP, which has low polarity (41), water causes a large shrinkage of silks by the destruction of hydrogen bonds in silk proteins (mainly amorphous regions), due to the high polarity (42) when the silks are immersed in water. We speculate that subsequent drying of the contracted silk fibers under axially restrained conditions, due to fixing the silk fibers on a paper clip (see table S1 for the details of sample preparation), leads to better alignment of the protein amorphous chains (43), resulting in decreased strain and slightly increased stress (44) (see result for SS/W in Fig. 2D).

We showed a large enhancement of the mechanical properties of spider silks via the MPI process and realized the prediction of Porter and Vollrath (45) that mineral-infused silk would show much greater toughness than native silk. This effect is not limited to spider silks but could also be shown to work for other biomaterials, such as collagen membranes from avian eggs (fig. S8).

References and Notes

- G. W. Bryan, P. E. Gibbs, *J. Mar. Biol. Assoc. U.K.* **59**, 969 (1979).
- P. E. Gibbs, G. W. Bryan, *J. Mar. Biol. Assoc. U.K.* **60**, 205 (1980).
- J. E. Hillerton, J. F. V. Vincent, *J. Exp. Biol.* **101**, 333 (1982).
- J. E. Hillerton, B. Robertson, J. F. V. Vincent, *J. Stored Prod. Res.* **20**, 133 (1984).
- R. Schofield, H. Lefèvre, *J. Exp. Biol.* **144**, 577 (1989).
- A. J. Edwards, J. D. Fawke, J. G. McClements, S. A. Smith, P. Wyeth, *Cell Biol. Int.* **17**, 697 (1993).
- D. L. J. Quicke, P. Wyeth, J. D. Fawke, H. H. Basibuyuk, J. F. V. Vincent, *Zool. J. Linn. Soc.* **124**, 387 (1998).
- H. C. Lichtenegger, T. Schöberl, M. H. Bartl, J. H. Waite, G. D. Stucky, *Science* **298**, 389 (2002).
- R. M. S. Schofield, M. H. Nesson, K. A. Richardson, *Naturwissenschaften* **89**, 579 (2002).
- H. C. Lichtenegger *et al.*, *Proc. Natl. Acad. Sci. U.S.A.* **100**, 9144 (2003).
- T. D. Morgan, P. Baker, K. J. Kramer, H. H. Basibuyuk, D. L. J. Quicke, *J. Stored Prod. Res.* **39**, 65 (2003).
- C. C. Broomell, M. A. Mattoni, F. W. Zok, J. H. Waite, *J. Exp. Biol.* **209**, 3219 (2006).
- M. G. Pontin, D. N. Moses, J. H. Waite, F. W. Zok, *Proc. Natl. Acad. Sci. U.S.A.* **104**, 13559 (2007).
- B. W. Cribb *et al.*, *Naturwissenschaften* **95**, 17 (2008).
- C. C. Broomell, F. W. Zok, J. H. Waite, *Acta Biomater.* **4**, 2045 (2008).
- F. Vollrath, D. P. Knight, *Nature* **410**, 541 (2001).
- A. Lazaris *et al.*, *Science* **295**, 472 (2002).
- Materials and methods are available as supporting material on Science Online.
- J. M. Gosline, P. A. Guerette, C. S. Ortlepp, K. N. Savage, *J. Exp. Biol.* **202**, 3295 (1999).
- J. C. Grosskreutz, *J. Electrochem. Soc.* **116**, 1232 (1969).
- In this experiment, we tried both a chamber-type ALD setup with variable precursor exposure times (the Savannah 100, Cambridge Nanotech) and a flow-type ALD setup (the SUNALE R75, Picosun, Finland) with negligible exposure times.
- We dipped the silk in solutions for 10 hours and compared the results with SS/ TiO_2 /300, the processing time of which also lasts ~10 hours. Because of the extreme volatility and hazard of the pure TMA or DEZ solution even in a glovebox, we were not able to perform the analogous experiment by dipping into TMA or DEZ.
- C.-Y. Chang, F.-Y. Tsai, S.-J. Jhuo, M.-J. Chen, *Org. Electron.* **9**, 667 (2008).

24. C. A. Wilson, R. K. Grubbs, S. M. George, *Chem. Mater.* **17**, 5625 (2005).
25. A. Fernández, L. R. Scott, *Phys. Rev. Lett.* **91**, 018102 (2003).
26. F. Cordier, S. Grzesiek, *J. Mol. Biol.* **317**, 739 (2002).
27. F. R. N. Gurd, P. E. Wilcox, *Adv. Protein Chem.* **11**, 311 (1956).
28. F. H. Arnold, B. L. Haymore, *Science* **252**, 1796 (1991).
29. J. T. Kellis, R. J. Todd, F. H. Arnold, *Nat. Biotechnol.* **9**, 994 (1991).
30. K. H. Gustavson, *Nature* **182**, 1125 (1958).
31. K. S. J. Rao, G. V. Rao, *Mol. Cell. Biochem.* **137**, 61 (1994).
32. H. Sun, H. Li, R. A. Weir, P. J. Sadler, *Angew. Chem. Int. Ed.* **37**, 1577 (1998).
33. T. Handel, W. F. DeGrado, *J. Am. Chem. Soc.* **112**, 6710 (1990).
34. Generally, in our ALD process, ZnO deposition has shown relatively poor self-limiting features. At longer exposure times, the process shows parasitic chemical vapor deposition (high film thickness with relatively poor uniformity). Therefore, in case of ZnO, an exposure of only 5 s was applied.
35. It has been observed that in the case of the metal-infiltrated silks, the contraction ratio produced by humidity up to 100% or temperature up to 70°C is in an almost negligible range (<3%).
36. A. H. Simmons, C. A. Michal, L. W. Jelinski, *Science* **271**, 84 (1996).
37. Y. Termonia, *Macromolecules* **27**, 7378 (1994).
38. N. Du et al., *Biophys. J.* **91**, 4528 (2006).
39. T. Lefèvre, M.-E. Rousseau, M. Pézolet, *Biophys. J.* **92**, 2885 (2007).
40. Y. Liu, Z. Shao, F. Vollrath, *Nat. Mater.* **4**, 901 (2005).
41. A. L. Let, D. E. Mainwaring, C. Rix, P. Murugaraj, *J. Non-Cryst. Solids* **354**, 1801 (2008).
42. Z. Shao, R. J. Young, F. Vollrath, *Int. J. Biol. Macromol.* **24**, 295 (1999).
43. R. E. Fornes, R. W. Work, N. Morosoff, *J. Polym. Sci. Polym. Phys. Ed.* **21**, 1163 (1983).
44. G. V. Guinea, M. Elices, J. Pérez-Rigueiro, G. R. Plaza, *J. Exp. Biol.* **208**, 25 (2005).
45. D. Porter, F. Vollrath, *Adv. Mater.* **21**, 487 (2009).
46. We thank W. Lee from the Korea Research Institute of Standards and Science in Daejeon, Korea; H. Han from the Department of Material Science and Engineering, Pohang University of Science and Technology in Pohang, Korea; and G.-M. Kim from the Department of Physics, Martin-Luther-University Halle-Wittenberg in Merseburg, Germany, for their helpful discussions on physical and chemical interpretations of mechanical properties, WAXS, NMR spectra, Raman spectra, energy-filtered TEM (EFTEM), and time-of-flight secondary ion mass spectroscopy (TOF-SIMS) data. We also thank J. Woltersdorf, M. Becker, and J. Liu from the Max Planck Institute of Microstructure Physics in Halle, Germany, for EFTEM and Raman measurements. We appreciate TOF-SIMS measurement by M. Krause and help and support in arranging the mechanical properties measurements and TOF-SIMS measurement by M. Petzold from the Fraunhofer Institute for Mechanics of Materials in Halle, Germany. This work has received financial support from the Bundesministerium für Bildung und Forschung (grant FKZ 03X5507) and Deutsche Forschungsgemeinschaft (grants BR 3370/3-1 and Mi 390/17-1).

Supporting Online Material

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Materials and Methods

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Figs. S1 to S11

Tables S1 to S4

References

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Asymmetric Autocatalysis Triggered by Carbon Isotope ($^{13}\text{C}/^{12}\text{C}$) Chirality

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Many apparently achiral organic molecules on Earth may be chiral because of random substitution of the 1.11% naturally abundant ^{13}C for ^{12}C in an enantiotopic moiety within the structure. However, chirality from this source is experimentally difficult to discern because of the very small difference between ^{13}C and ^{12}C . We have demonstrated that this small difference can be amplified to an easily seen experimental outcome using asymmetric autocatalysis. In the reaction between pyrimidine-5-carbaldehyde and diisopropylzinc, addition of chiral molecules in large enantiomeric excess that are, however, chiral only by virtue of isotope substitution causes a slight enantiomeric excess in the zinc alkoxide of the produced pyrimidyl alkanol. Asymmetric autocatalysis then leads to pyrimidyl alcohol with a large enantiomeric excess. The sense of enantiomeric excess of the product alcohol varies consistently with the sense of the excess enantiomer of the carbon isotopically chiral compound.

The limits of knowledge in recognizing and understanding chirality are continually pushed back by innovative methods (1–4). Although the theory of a tetrahedral asymmetric carbon atom with four differing substituents has been understood since the 19th-century work of van't Hoff and LeBel (5–9), molecules for which this characteristic arises solely from carbon isotopic substitution are rare (10). It is not known whether a reactive difference between enantiomers can arise solely from a difference in carbon isotopic substitution. Such an observation has been experimentally inaccessible. Considering that the natural abundance of ^{13}C is ~1.11% on Earth, many organic molecules may be chiral arising from carbon isotope differences in enantiotopic

groups, although the enantiomeric excess arising from this source may be minute (Fig. 1).

There are reports on enantioselective reactions (11, 12) or on a kinetic resolution (13) induced by hydrogen isotope (H/D) replacement, and the use of hydrogen isotopes to explore enzymatic mechanisms is well known (14). In a polymer, isotope chirality (H/D) has been shown to cooperatively control macromolecular helical handedness (15). However, the effect of carbon isotope substitution is expected to be far smaller. Thus, enantioselective reactions induced by molecules that are chiral

solely by virtue of carbon isotope substitution present a challenging problem. To the best of our knowledge, there have been no reports in which carbon isotopically substituted chiral compound induces enantioenrichment in an asymmetric reaction.

Here, we show that asymmetric autocatalysis with amplification of enantioenrichment (16, 17) can be successfully applied to the discrimination between enantiomers that differ solely in substitution of carbon isotopes. Asymmetric autocatalysis has enormous power to recognize molecular chirality (18–22). In the reaction between pyrimidine-5-carbaldehyde **4** and diisopropylzinc ($i\text{-Pr}_2\text{Zn}$), the zinc alkoxides of the isotopically substituted chiral alcohols **1** to **3** cause a slight enantiomeric excess (ee) in the zinc alkoxide of pyrimidyl alkanol **5**. After autocatalytic amplification of this enantiomeric excess, alkanol **5** accumulates an enhanced ee with a sense that varies consistently with that of the abundant enantiomer of the isotopically substituted chiral compound **1**, **2**, or **3** (Fig. 2).

The isotopically chiral compounds **1** to **3** were synthesized as shown in Fig. 3 (23). First, methyl- ^{13}C -methylphenyl methanol **1** was prepared. To restrict chiral effects to the influence of the carbon isotopic chirality, both enantiomers of **1** were prepared from the same chiral ligand (–)-**8** as shown in Fig. 3, route I. The enantiomers could be synthesized by direct asymmetric dimethylzinc addition to acetophenone using sulfonamide (–)-**8** (24) as a chiral ligand. (R)-Alcohol **1** was synthesized using ^{13}C -labeled acetophenone **6** as the substrate. On the other hand, (S)-**1** was

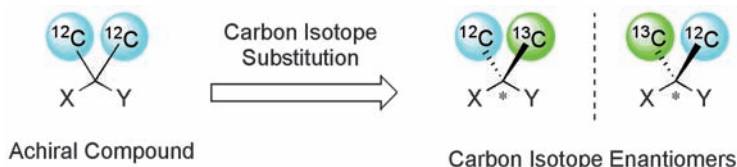


Fig. 1. Generation of chirality by carbon isotope substitution.

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Greatly Increased Toughness of Infiltrated Spider Silk

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Improving on Spider Silk?

In a conventional atomic layer deposition, alternating pulses of a reactive metal-based precursor and a reactant, such as water, are deposited onto a solid surface to create a coating that conforms closely to the original surface. **Lee et al.** (p. 488) apply this process to fibers of spider silk, which are soft and porous. In addition to forming a coating on the fibers, some of the metal ions penetrate the fibers and react with the protein structure. These doped fibers show significant increase in toughness as measured by the work required to deform them.

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